pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Duration and DV01

WEEK 2 · DAY 4

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From price to price risk

If yields move 1 bp, what does each position make or lose?

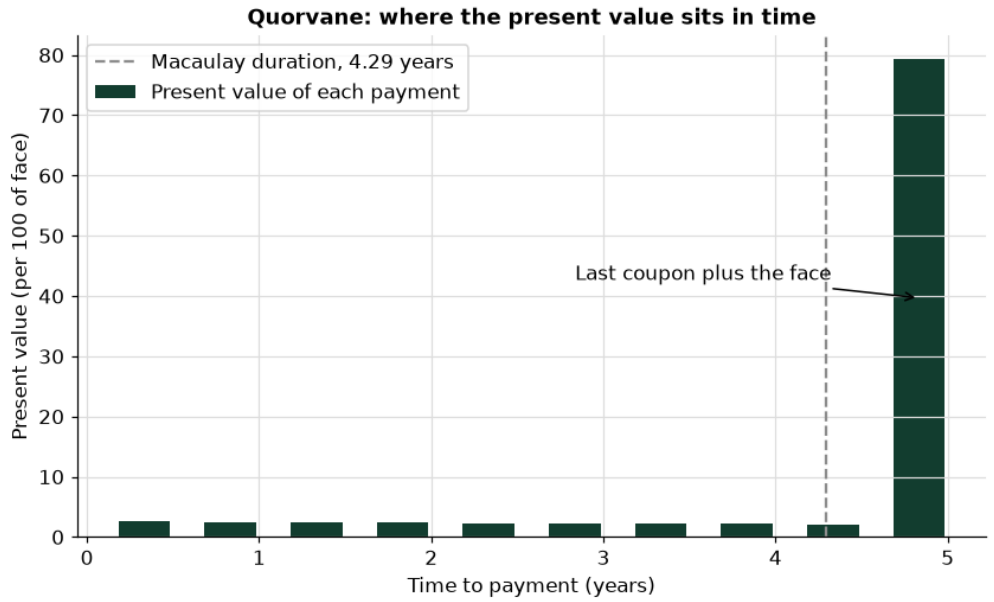
Issuer	Par held	Dirty price
Quorvane Energy	5,000,000	100.236551
Dunmarrow Rail	3,000,000	101.010586
Pellwick Paper	4,000,000	101.179914
Halvering Foods	2,000,000	99.872491

Settled Monday 30 November 2026, at day 8's prices. The positions are invented.

Week 2: dates, accrued, and risk

Day 6	Dates and coupon schedules
Day 7	Day counts and accrued interest
Day 8	Clean and dirty price
Day 9	**Duration and DV01**
Day 10	Convexity, the September book

Macaulay duration: the balance point in time



Each payment's time, weighted by its present value over the dirty price.

4.2932

years: Quorvane's Macaulay duration

4.8333

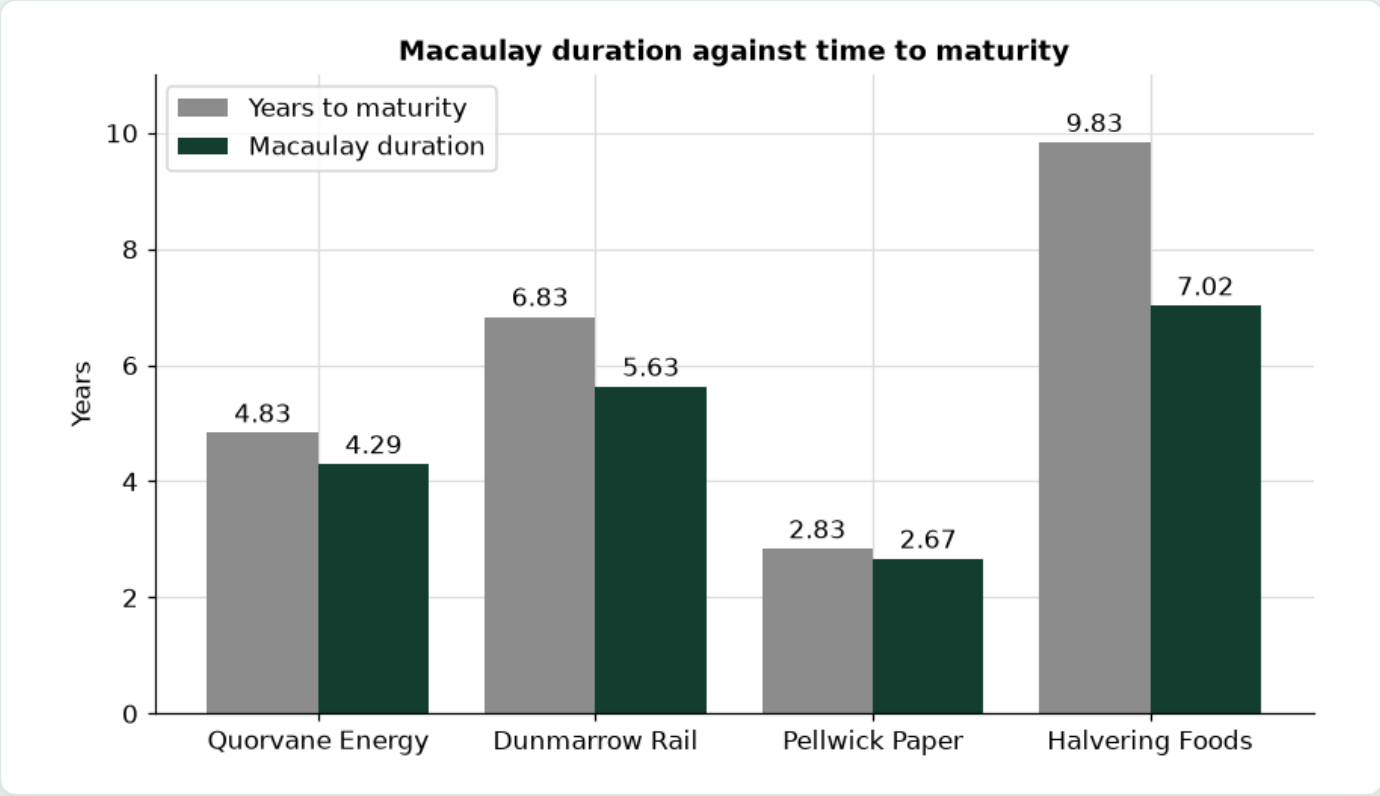
years to the final payment

Macaulay, NBER, 1938: duration is "a reality of which 'maturity' is only one factor".

Excel side by side (tested in Excel)

`=DURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)`	4.293248
`=DURATION(DATE(2026,11,30),DATE(2031,9,30),0,5.4/100,2,0)`	4.833333

Duration against maturity, drawn



Every duration is shorter than its maturity: the coupons arrive first and pull the average in.

The gap widens with maturity: the longest bond sits furthest below its maturity.

Modified duration: price sensitivity

```
modified = Macaulay / (1 + y / 2)
```

The price falls about modified duration percent for each 100 bp rise in yield.

Typed out, the division returns MDURATION to the last digit. The /2 is the frequency.

Quorvane, dirty	Repriced	Duration line
+1 bp, percent	-0.04179333	-0.04180378
+100 bp, per 100	-4.087492	-4.190266
-100 bp, per 100	+4.296930	+4.190266

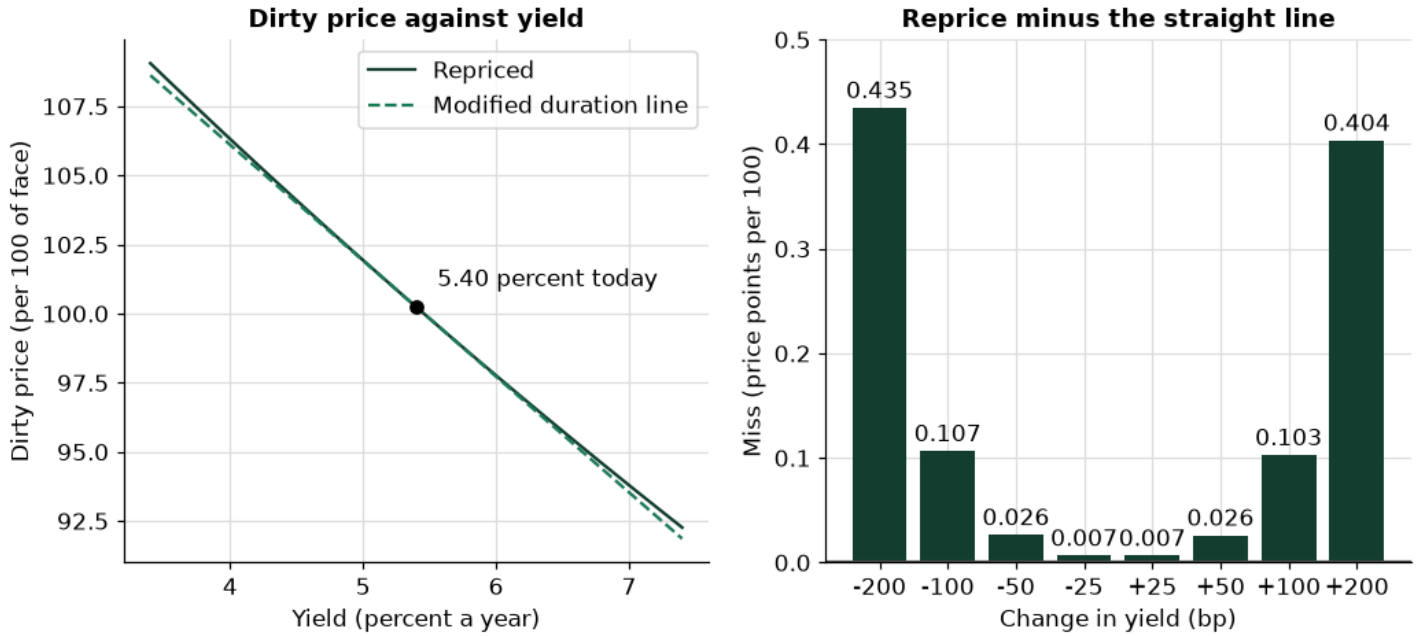
Big moves: above the line, both ways.

Duration is the slope at today's yield, and the curve bends. Day 10 measures the bend: convexity.

Excel side by side (tested in Excel)

`=MDURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)`	4.180378
`=DURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)/(1+5.4/100/2)`	4.180378

Duration's straight line, drawn



The line touches the curve at today's yield and the curve bends above it. Every miss is positive and grows with the move.

A fall in yield misses by more than a rise of the same size. Day 10 measures the bend.

DV01 per 100 and per position

```
DV01 = modified x dirty price x 0.0001
```

Issuer	DV01 per 100	Position, dollars
Quorvane Energy	0.041903	2,095.13
Dunmarrow Rail	0.055175	1,655.26
Pellwick Paper	0.026359	1,054.35
Halvering Foods	0.067511	1,350.23
Book		6,154.97

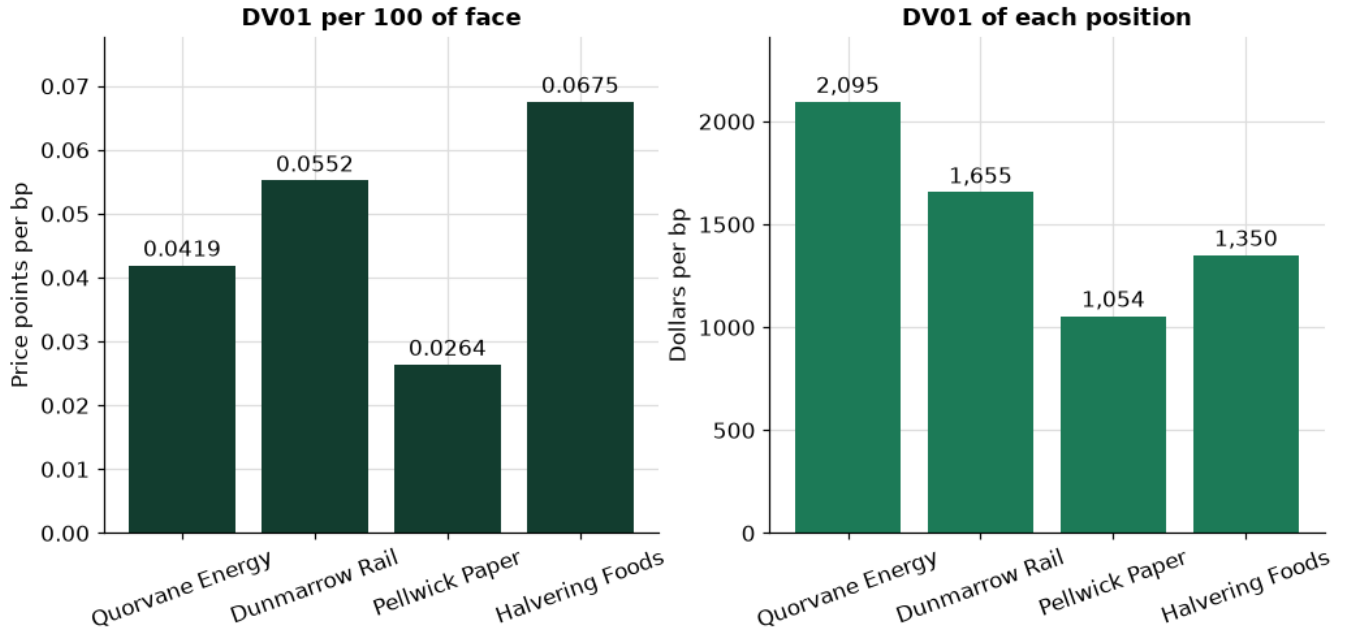
Position DV01 in dollars = DV01 x par held / 100. Dollars add up across positions.

Excel side by side (tested in Excel)

B1	`=PRICE(...)`
B2	`=100*5.25/100/2*COUPDAYBS/COUPDAYS`
B4	`=MDURATION(...)`
B5	`=B4*(B1+B2)/10000`
B11	`=B5*5000000/100`

B5 returns 0.041903; B11 returns 2,095.13.

DV01 per 100 and per position, drawn



The panels rank the bonds differently. Per 100, the longest bond carries the most. In dollars, the largest position does.

Check DV01 by bumping the yield

```
down = bondmath.price(s, m, 5.25, 5.4 - 0.01)
up = bondmath.price(s, m, 5.25, 5.4 + 0.01)
(down - up) / 2      # half the 2 bp change
```

Clean prices are fine: accrued does not move with yield, so it cancels in the difference.

Not for a bond in its final period: Excel's PRICE uses simple interest there.

Excel side by side (tested in Excel)

B8	`=PRICE(...,(5.4-0.01)/100,...)`	99.403465
B9	`=PRICE(...,(5.4+0.01)/100,...)`	99.319659
B10	`=(B8-B9)/2`	0.041903

1.9e-9

per 100: B10 minus the DV01 formula in B5

Signs: what a positive DV01 means

Change in value for a move of n bp = minus position DV01 x n

Long: 5,000,000 Quorvane

Position DV01: +2,095.13 dollars

Yield 1 bp lower: gains about that much

Yield 1 bp higher: loses about that much

Short: -2,000,000 Halvering

Position DV01: -1,350.23 dollars

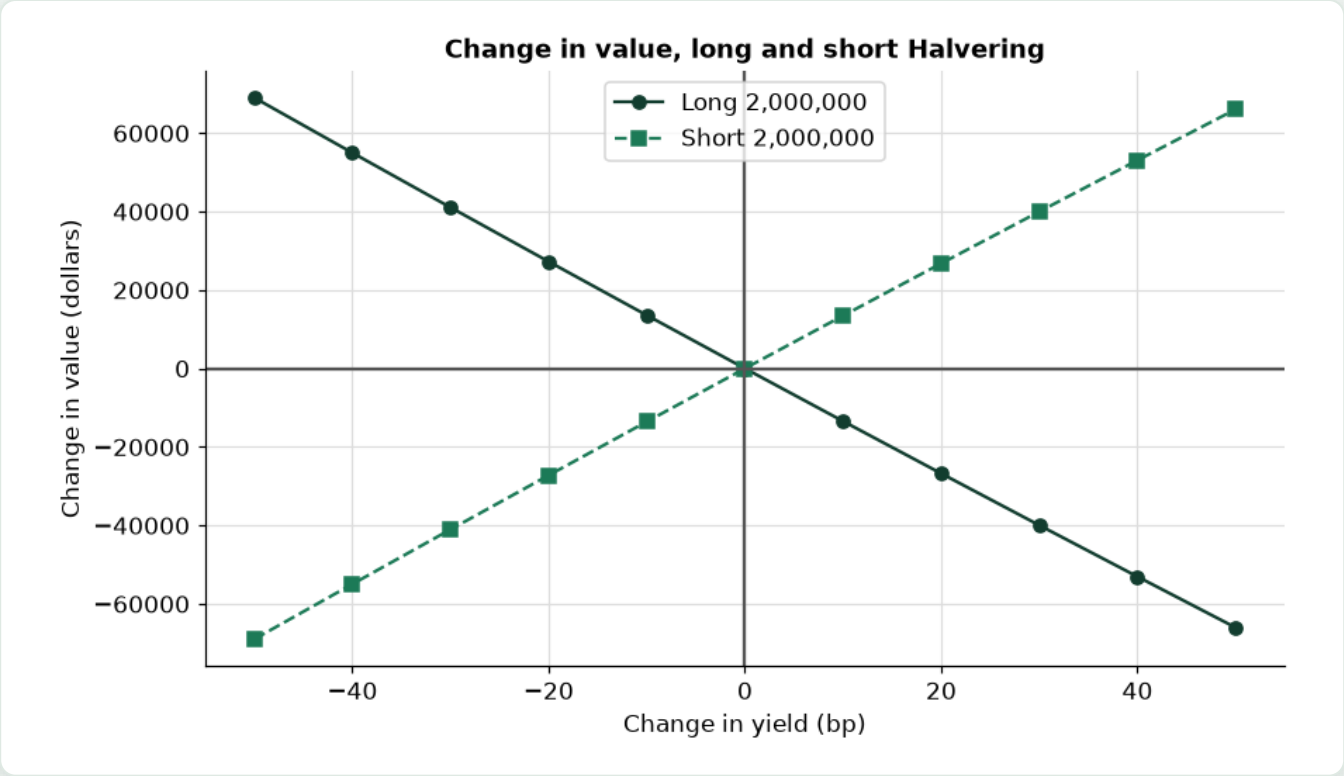
Yield 1 bp lower: loses about that much

Yield 1 bp higher: gains about that much

This course reports DV01 positive for a long. Some systems flip it: check before adding two sources.

Arithmetic on signs, not a view on where yields go.

Long and short, drawn



The long position loses when the yield rises. The short is its mirror image. The slope at zero is the position DV01.

The holdings duration column, reproduced

200

bonds in holdings.csv, as of 30 September 2026

0.456824

years: Macaulay's largest miss against the file

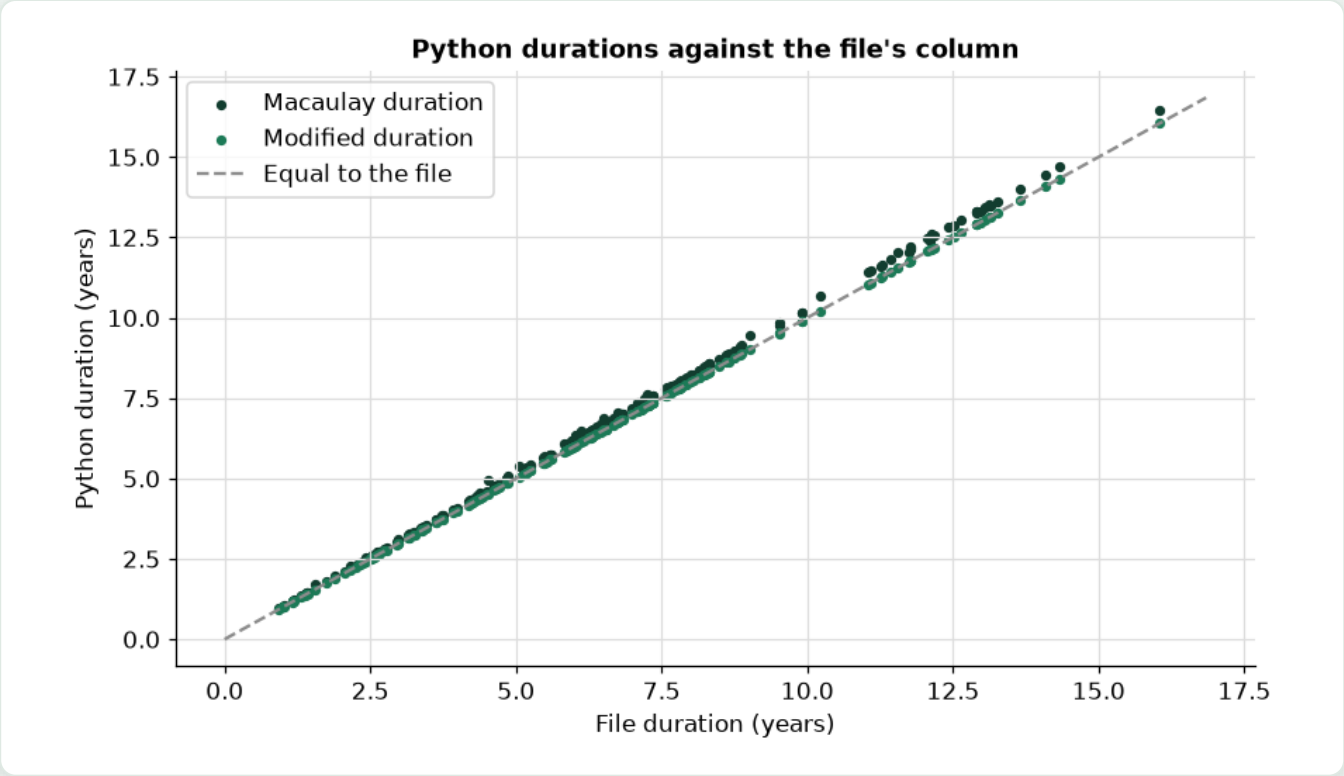
0.000498

years: modified duration's largest gap, within rounding

The file's duration column is modified duration: price sensitivity.

```
holdings["modified_python"] = [bondmath.modified_duration(*bond) for bond in inputs]
assert holdings["gap"].max() <= 0.0005 + 1e-9   # the file rounds to 3 decimals
```

The holdings column, drawn



Modified duration sits on the line. Macaulay sits above it, further for longer bonds: the file's column is modified duration.

Today's Excel, every formula tested

Job	Excel (tested), Quorvane	Returns
Macaulay	`=DURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)`	4.293248
Zero coupon	`=DURATION(DATE(2026,11,30),DATE(2031,9,30),0,5.4/100,2,0)`	4.833333
Modified	`=MDURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)`	4.180378
DV01 per 100	`=B4*(B1+B2)/10000`	0.041903
Position	`=B5*5000000/100`	2,095.13
The bump	`=(B8-B9)/2`	0.041903

B1 PRICE, B2 the accrued formula, B4 MDURATION; B8 and B9 PRICE 1 bp either side.

Look back: a file's history

git log --oneline -- bondmath.py

```
ef5648a Add macaulay_duration, ...
e722e2c Start of day 9: ...
```

Only the commits that touched bondmath.py. Everything after -- is a file name.

git diff HEAD~1 -- bondmath.py

```
+def macaulay_duration(settlement: ...
+def modified_duration(settlement: ...
+def dv01(settlement: date, ...
```

HEAD is the last commit, HEAD~1 the one before. Every line is +: added, nothing taken out.

A test that passed yesterday fails today? The diff since yesterday is the first place to look.

Hashes differ on every run.

Bring it home: paste, test, commit, look back

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Three functions and four tests at the end of your files, without the %%writefile line.

3

Test

python -m pytest -q: 27 passed, or 35 with the exercises of days 1 to 8.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Look back

git log --oneline -- bondmath.py: a commit for every day you brought home.

```
git diff HEAD~1 -- bondmath.py
```

In Colab: download bondmath.py and test_bondmath.py from the work folder under /tmp. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert against Excel's reference rows. Run it: it fails.

→

4

Fix

Find the convention it broke, fix it, run the test again.

Today: modified duration divides by (1 + y), the whole annual yield, instead of (1 + y / 2).

Code that runs is not the same as code that is right.

What you can do now

- Compute Macaulay duration and match Excel's DURATION
- Turn it into modified duration and match MDURATION
- Compute DV01 per 100 and per position, and a book DV01
- Check DV01 by bumping the yield; read its sign, long and short
- Reproduce the holdings duration column; look back with git log and git diff

Week 2, Day 5 Convexity, where duration's straight line misses, and the September book